

2.

E) We need to discover a code that spans the correctable set = $\text{span}\{I, X, X^{-1}, Z, Z^{-1}\} = \text{span}(E_n)$
 Since all these errors need to be non-degenerate

$$|\psi\rangle = \alpha|0_L\rangle + \beta|1_L\rangle$$

$$\dim(C) = 2 = \dim(IC)$$

$$\dim(X) = 2 ; \dim(Z) = 2$$

$$\dim(X^{-1}) = 2 ; \dim(Z^{-1}) = 2.$$

Every error will be in a subspace that is orthogonal to all the other subspaces.

So, $5 \times 2 = 10$ dimensions (at least, for all errors to be non degenerate)

$$\therefore q \geq 10$$

... QED.